



Mars 2026

Tutorial: Visual Simulator Module

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1. Introduction

The Visual Simulator module (also called Configuration Viewer) is an animated interface that visually presents the effects of Ortho4XP V3 configuration parameters.

In this interface you see in real time what each slider does, thanks to an animated canvas and help thumbnails. The animations let you understand immediately what will happen when the tile is built.

Recommended settings are shown under each slider. Explanations appear in the language chosen at launch (French or English).

This module does not replace the Config window: it illustrates it. Displayed values come from the active tile's .cfg (or global defaults). You can change them and write them back to the configuration file.

Step-by-step workflow

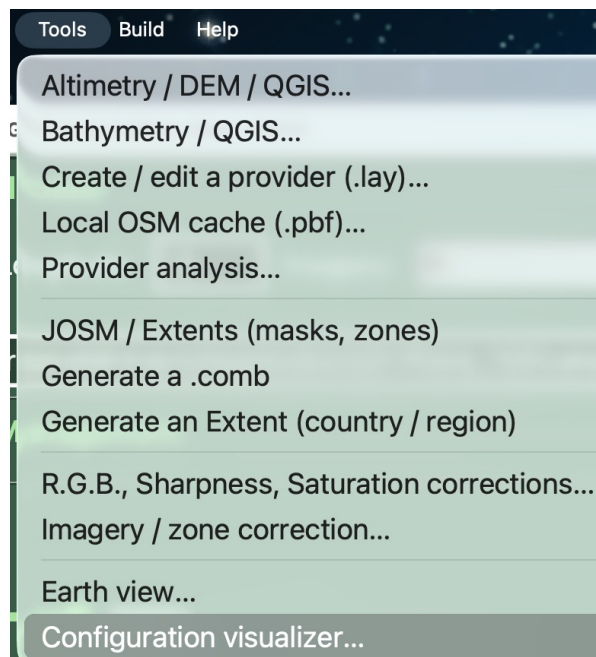
1) Step-by-step workflow

- Open Ortho4XP V3 and select (or create) the active tile.
- Tools menu → Configuration Viewer...
- Check that the message "Values loaded from cfg" appears.
- Browse the 4 tabs. Move the sliders and watch the canvas + thumbnails.
- Adjust according to the recommendations shown under each control.
- Click "Write tile cfg" to apply to the current tile (or "Write app cfg" for default values).
- Close the simulator and start the tile build (Assemble Vector data → ... → Build Imagery/DSF).
- After a first test flight, return to the simulator to fine-tune if needed.

2. Accessing the module

In the main Ortho4XP V3 menu bar:

- Tools menu → Configuration Viewer...

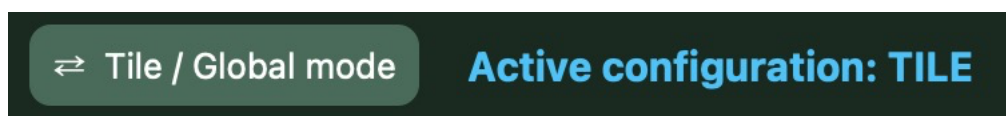


General interface

Tile / Global mode and Simulator buttons

The top bar of the Simulator shows **which configuration is active: TILE or GLOBAL**.

- ⇌ **Tile / Global mode**
- Switches between the two settings sources, like **Tools → Config**.
- **TILE** = settings for the current tile (Ortho4XP_±lat±lon.cfg).



- **GLOBAL** = application defaults (Ortho4XP.cfg). One click changes the mode **and** reloads the sliders from the chosen source.



The 4 DEM sections

custom_dem	/Volumes/SSD/Altimétries Originaux/Altimétrie assemble/Assemble France/+46-003/+46-003.tif	
custom_bathy_dem		
custom_scenery_dir		
custom_overlay_src		

1. dem_sonny_dir

Folder for the **Sonny altimetry structure** (data organised by country).

Where Ortho4XP finds / manages the current Sonny elevation tree. Set this if you use the V3 Sonny-based altimetry layout.

2. dem_asc_dir

Folder for **raw ASC sources** (original elevation files, not yet prepared for the tile). Entry point for “as-is” elevation rasters before reduction / reprojection.

3. dem_epsg_dir

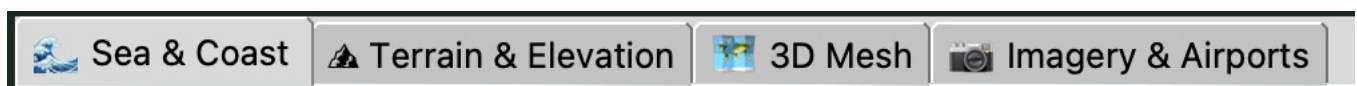
Folder for **reduced EPSG output**: elevation already processed into geographic coordinates for the Ortho4XP pipeline. Intermediate step between raw ASC and per-tile assembly.

4. dem_tile_out_dir

Folder for **per-tile assembly output**: final prepared

altimetry for a given tile. Typically what feeds the tile mesh build (or what you point to via a custom DEM, depending on your workflow).

The tabs:



➤ Top tab bar: Sea & Coast · Terrain & Relief · Mesh 3D · Imagery & Airports

➤ Animated canvas (visual area) at the top of each tab: shows the effect of settings in real time

➤ Groups of sliders and drop-down lists in the centre, with an explanation under each control

➤ Mini-animation thumbnails to the right of some sliders

➤ Button bar at the bottom of the window

↶ Reload from Tile config

↶ Reload from Global config

✓ Write Tile cfg

🌐 Write App cfg

✖ Close

➤ **↶ Reload from Tile config** Reads the **tile** cfg file and updates all sliders (does not write).



➤ **↶ Reload from Global config** Reads the **global** cfg file and updates all sliders (does not write).



➤ **✓ Write Tile cfg** Saves current slider values into the **tile** cfg. Other keys in the file (e.g. language=, V3 DEM folders) are **kept**.



➤ **🌐 Write App cfg** Saves current values into the **global** cfg (Ortho4XP.cfg), also **preserving** keys not managed by the Simulator.



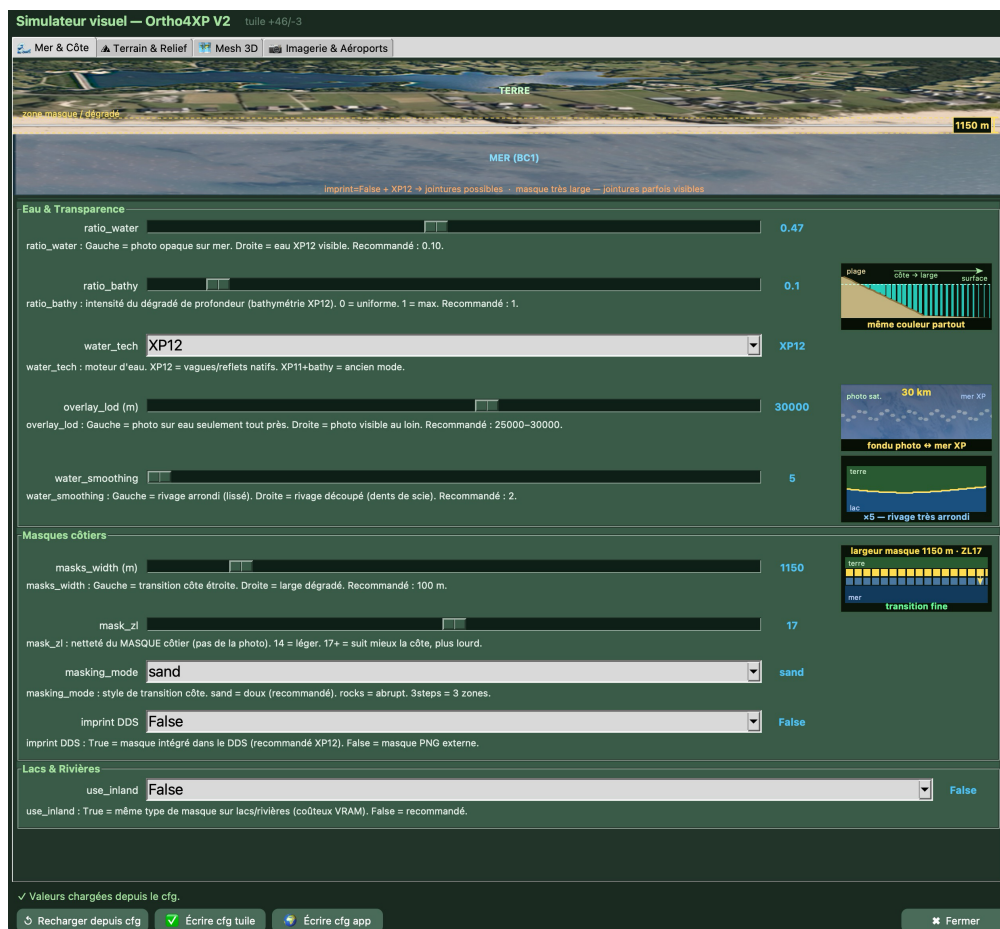
➤ **✖ Close** Closes the Simulator window (nothing is saved automatically).

Tip: Check *Active configuration* first, reload the source you want, adjust sliders while watching the vignettes, then **Write** Tile or App as needed.

3. Sea & Coast tab

T

his tab controls water appearance, the coast/sea transition and the transparency masks required by X-Plane 12.



3.1 Water & Transparency group



Parameter	Function / Slider direction	Recommended »
ratio_water	Left = opaque photo over sea. Right = XP12 water visible (transparency).	≈ 0.10
ratio_bathy	Strength of the depth colour gradient (XP12 bathymetry). 0 = uniform, 1 = max.	1
water_tech	Water engine: XP12 = native waves/reflections. XP11+bathy = legacy mode.	XP12
overlay_lod (m)	Left = photo on water only nearby. Right = photo visible far away.	25 000-30 000
water_smoothing	Left = rounded shoreline (smoothed). Right = jagged shoreline (saw-tooth).	≈ 2

3.2 Coastal masks group

Coastal Masks

masks_width (m)

1150

masks_width: Left = narrow shore blend. Right = wide blend. Recommended: 100 m.

mask_zl

17

mask_zl: coastal MASK resolution (not the photo). 14 = light. 17+ = follows coast better, heavier.

masking_mode

sand

sand

masking_mode: shore blend style. sand = soft (recommended). rocks = abrupt. 3steps = 3 zones.

imprint DDS

False

False

imprint DDS: True = mask baked into DDS (recommended for XP12). False = external PNG mask.

largeur masque 1150 m - ZL17

terre

mer

transition fine

Parameter	Function / Slider direction	Recommened
masks_width (m)	Left = narrow shore transition. Right = wide blend.	≈ 100 m
mask_zl	Sharpness of the coastal MASK (not the photo). 14 = light, 17+ = follows coast better (heavier).	16-17
masking_mode	Transition style: sand = soft, rocks = abrupt, 3steps = 3 zones.	sand
imprint DDS	True = mask baked into DDS (recommended for XP12). False = external PNG mask.	True (XP12)

3.3 Lakes & Rivers group

Lakes & Rivers

use_inland

False

False

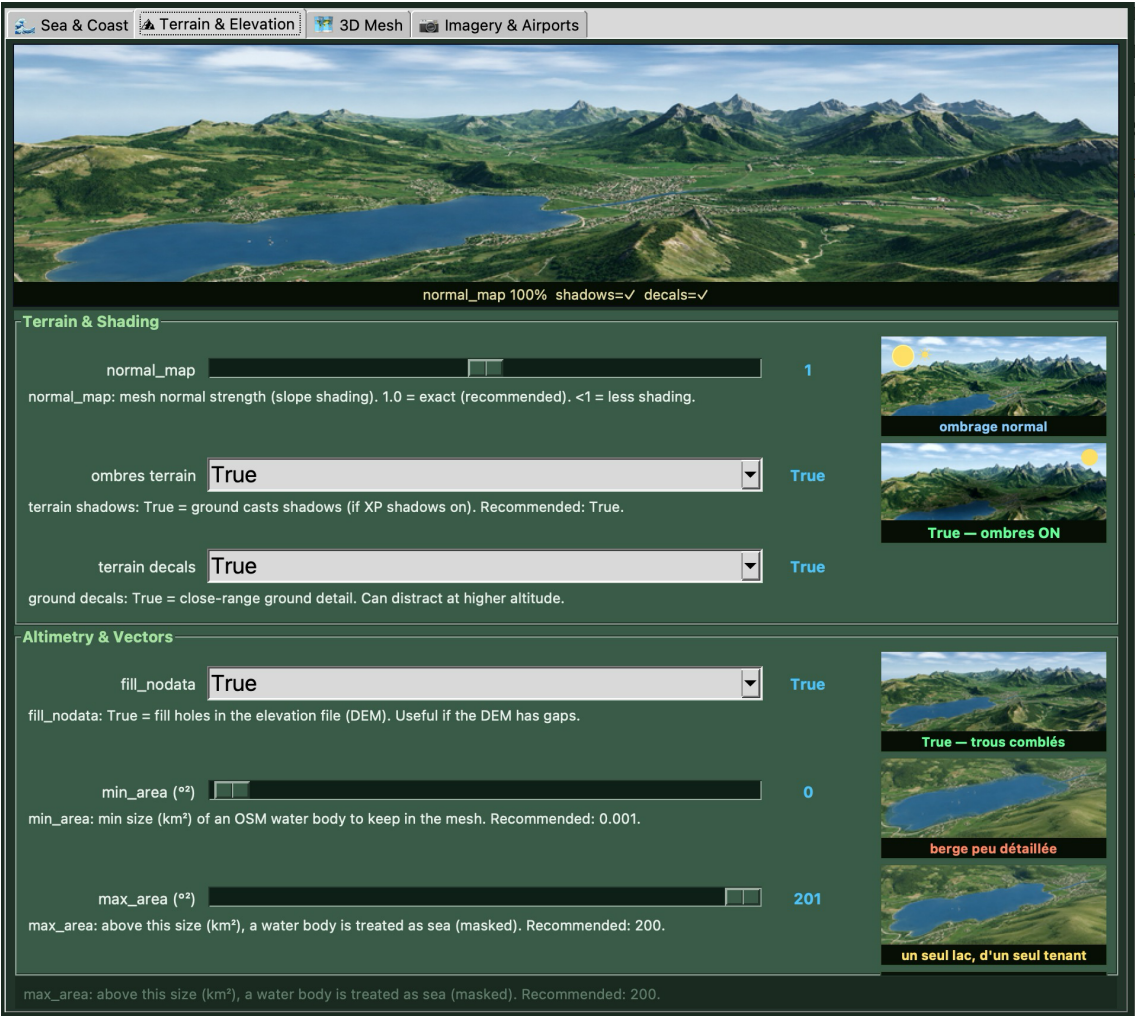
use_inland: True = same masks on lakes/rivers (VRAM heavy). False = recommended.

Parameter	Function / Slider direction	Recommened
use_inland	True = same mask type on lakes/rivers (VRAM-heavy). False = recommended.	False

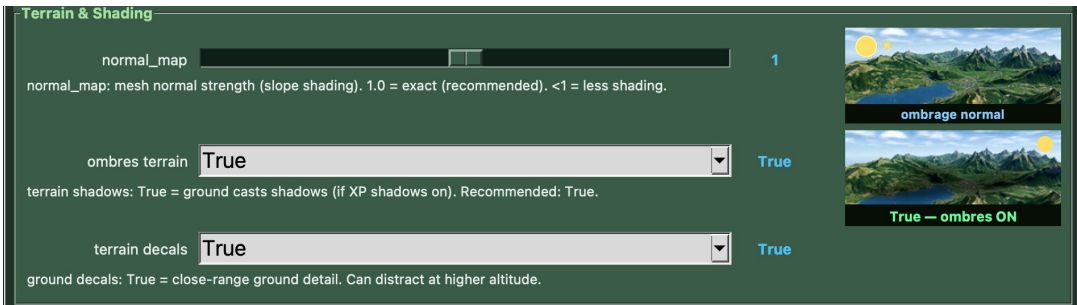
The top canvas shows in real time the mask width (yellow dashed line) and the transition type. Thumbnails on the right illustrate ratio_bathy, overlay_lod and water_smoothing.

4. Terrain & Relief tab

Controls ground shading, detail decals, DEM hole filling and OSM water-body handling.



4.1 Terrain & Shading group



Parameter	Function / Slider direction	Recommended
normal_map	Mesh normal strength (slope shading). 1.0 = exact. < 1 = less shading.	1.0
terrain shadows	True = ground casts shadows (if XP shadows are enabled).	True
terrain decals	True = close-range ground detail texture. Can distract at higher altitude.	True or False

4.2 Altimetry & Vectors group

Altimetry & Vectors

fill_nodata

True

True

fill_nodata: True = fill holes in the elevation file (DEM). Useful if the DEM has gaps.

min_area (°²)

0

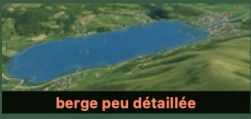
min_area: min size (km²) of an OSM water body to keep in the mesh. Recommended: 0.001.

max_area (°²)

201

max_area: above this size (km²), a water body is treated as sea (masked). Recommended: 200.







Parameter	Function / Slider direction	Recommen ded
fill_nodata	True = fill holes in the elevation file (DEM). Useful if the DEM has gaps.	True
min_area (°²)	Minimum size (km²) of an OSM water body to keep in the mesh.	≈ 0.001
max_area (°²)	Above this size (km²) a water body is treated as sea (masked).	≈ 200
water_simpl	Left = simplified shore (negative). Right = detailed shore (positive).	as needed

5. Mesh 3D tab

3D mesh triangulation settings: density, curvature tolerance, triangle budget and geometry cleanup.

Sea & CoastTerrain & Elevation3D MeshImagery & Airports



mesh_zl 19 | curvature → mixte | curvature → partiel | limit → peu de Δ

Mesh Parameters

mesh_zl

19

mesh_zl: max ZL the mesh is built for. Must be ≥ your photo ZL. Recommended: 19.

curvature_tol

16

curvature_tol: Left = large triangles, coarse relief. Right = small triangles, precise relief.

limit_tris (M)

15

limit_tris: Left = few triangles (incomplete mesh). Right = large budget. Recommended: 1–3 M.

min_angle (°)

2

min_angle: min triangle angle (°). Avoids skinny triangles. Recommended: ~10.

Quality & Cleanup

iterate

0

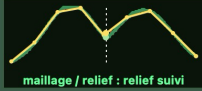
iterate: Left = 1 fast pass (coarse coast). Right = more passes (fine coast, slower).

clean_geom

True

True

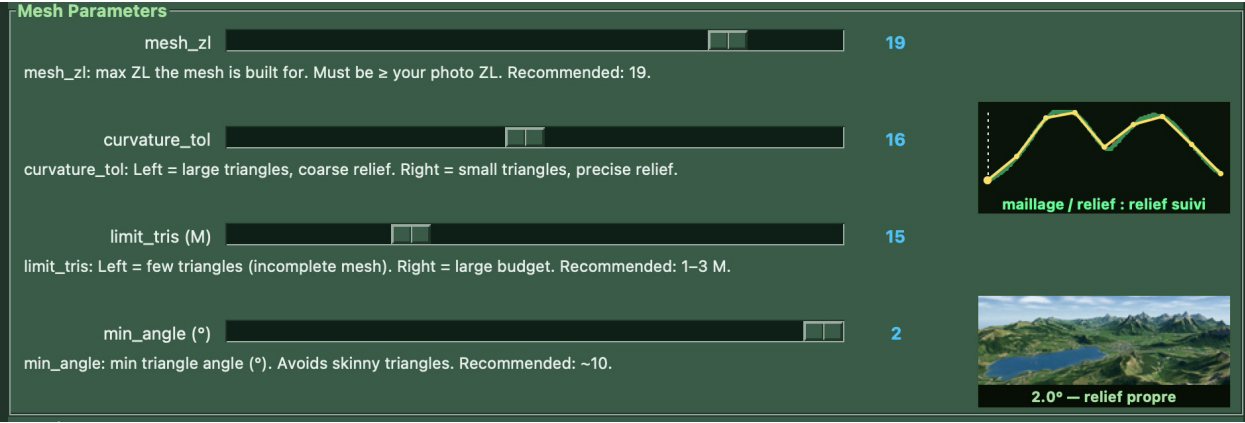
clean_bad_geometries: True = fix invalid OSM geometries before meshing. Recommended: True.





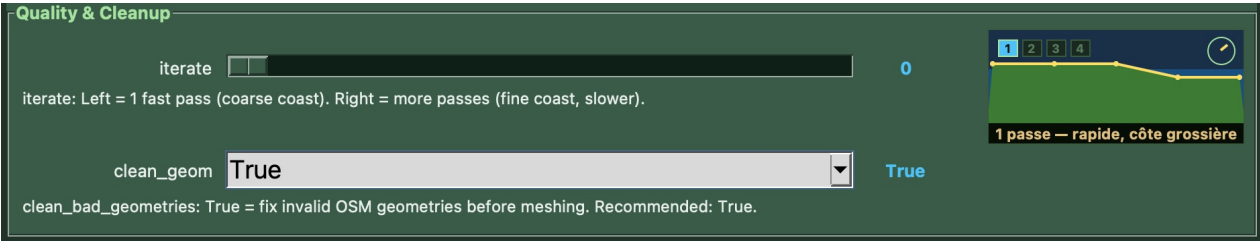


5.1 Mesh parameters group



Parameter	Function / Slider direction	Recommended
mesh_zl	Max ZL the mesh is prepared for. Must be $\geq$ photo ZL.	19
curvature_tol	Left = large triangles, coarse relief. Right = small triangles, precise relief.	depends on terrain
limit_tris (M)	Left = few triangles (incomplete mesh). Right = large budget.	1-3 M
min_angle (°)	Minimum triangle angle. Avoids overly skinny triangles.	≈ 10

5.2 Quality & Cleanup group

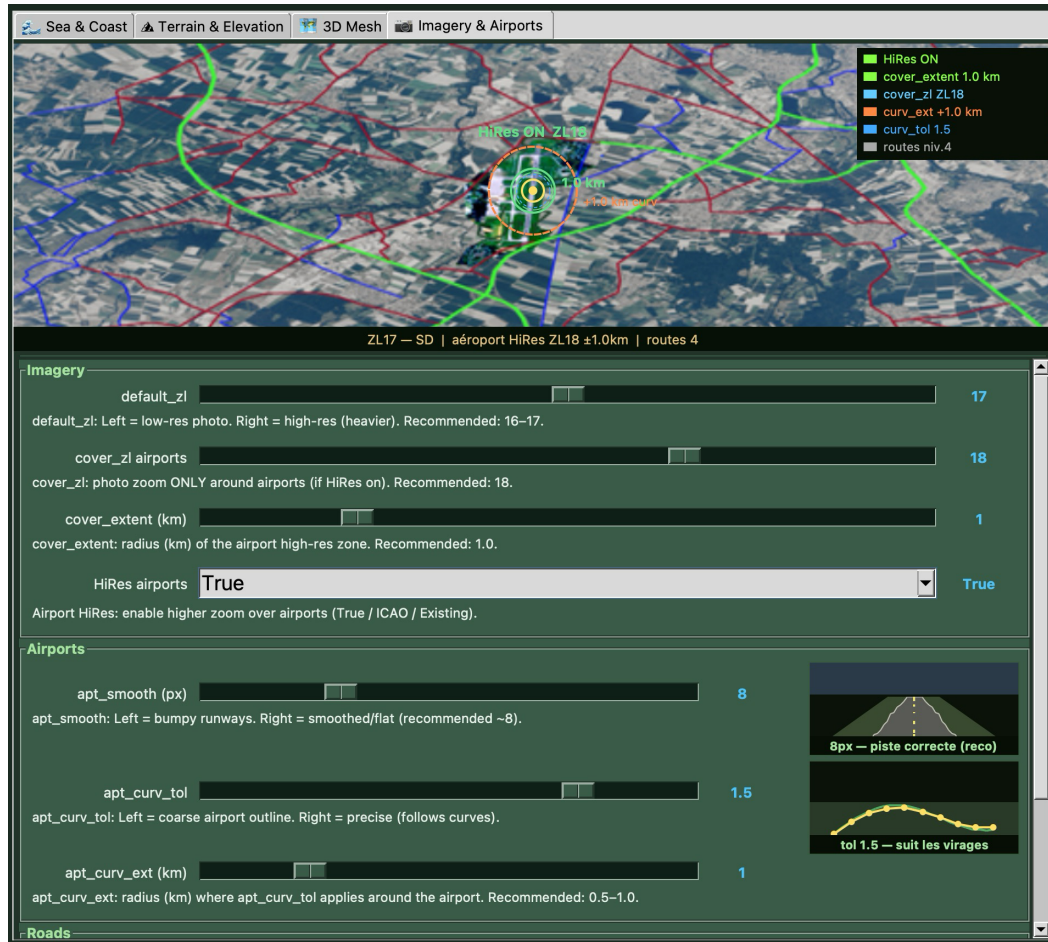


Parameter	Function / Slider direction	Recommended
iterate	Left = 1 fast pass (coarse coast). Right = more passes (fine coast, slower).	0-1
clean_geom	True = repair invalid OSM geometries before meshing.	True

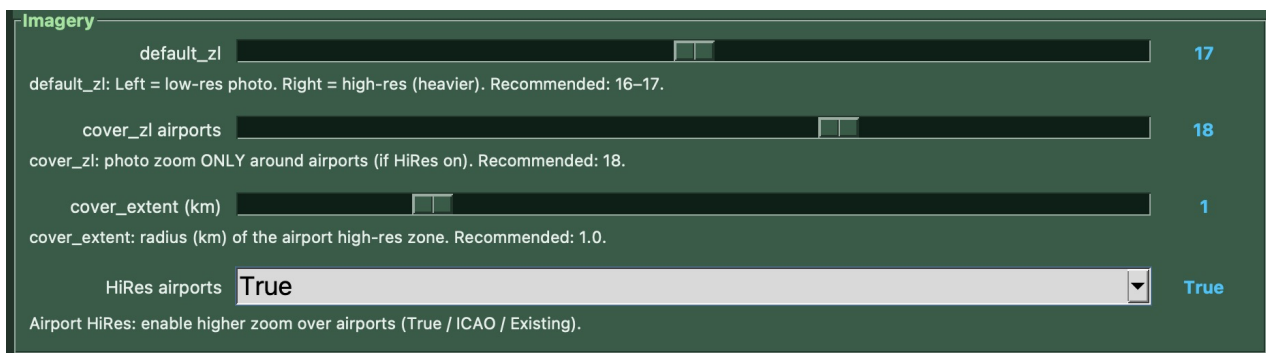
The canvas shows a 3D landscape whose mesh reacts to `curvature_tol` and `limit_tris`. Thumbnails illustrate relief following and the number of passes.

6. Imagery & Airports tab

Orthophoto zoom settings, high-resolution zones around airports, runway smoothing and road levelling.



6.1 Imagery group



Parameter	Function / Slider direction	Recommended
default_zl	Left = low-definition photo. Right = high definition (heavier).	16-17
cover_zl airports	Photo zoom ONLY around airports (when HiRes is enabled).	18
cover_extent (km)	Radius (km) of the high-resolution airport zone.	≈ 1.0
HiRes airports	Enable higher zoom over airports: False / True / ICAO / Existing.	True or ICAO

6.2 Airports group

Airports

apt_smooth (px)

8

apt_smooth: Left = bumpy runways. Right = smoothed/flat (recommended ~8).

apt_curv_tol

1.5

apt_curv_tol: Left = coarse airport outline. Right = precise (follows curves).

apt_curv_ext (km)

1

apt_curv_ext: radius (km) where apt_curv_tol applies around the airport. Recommended: 0.5-1.0.

8px — piste correcte (reco)

tol 1.5 — suit les virages

Parameter	Function / Slider direction	Recommended
apt_smooth (px)	Left = bumpy runways. Right = smoothed/flat runways.	≈ 8
apt_curv_tol	Left = coarse airport outline. Right = precise outline (follows curves).	1.0-1.5
apt_curv_ext (km)	Radius (km) where apt_curv_tol applies around the airport.	0.5-1.0

6.3 Roads

Roads

road_level

4

road_level: Left = no roads levelled. Right = all road types levelled (0→5).

levelled_segs

200000

levelled_segs: Left = bumpy roads (roller-coaster). Right = flattened roads (fixed).

X

X

200k — en partie nivelé

Parameter	Function / Slider direction	Recommended
road_level	Left = no roads levelled. Right = all road categories (0→5).	3-4
max_levelled_segs	Left = bumpy roads (roller-coaster). Right = flattened roads.	depends on region

The canvas shows an aerial view with HiRes zones (green/orange circles) and the road network coloured by levelling level.

7. Practical tips

- General slider direction: left = “raw” / less favourable effect; right = correction / more detail (except documented cases).
- For X-Plane 12: prefer `water_tech = XP12`, `imprint DDS = True`, `ratio_water ≈ 0.10`.
- Do not raise `mask_zl` or `mesh_zl` unnecessarily: build time and file size increase quickly.
- On mountainous terrain, lower `curvature_tol` a little (more triangles) and increase `limit_tris`.
- Airport HiRes zones (`cover_zl 18 + cover_extent 1 km`) give an excellent quality/size ratio.
- After changes, remember to write the `cfg` before rebuilding the tile.

8. Summary

The Visual Simulator turns technical parameters into understandable images and animations. Each tab covers one domain (water, terrain, mesh, imagery/airports).

The bottom buttons synchronise values with the configuration files.

Use it systematically before starting a long build: you will save time and avoid unnecessary rebuilds.

Yellow dashed boxes are image placeholders — insert additional detail screenshots if desired.